E-COMMERCE PLATFORM FOR PERSONALIZED RETAIL EXPERIENCES: SINHALA VOICE ASSISTANT

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Dissertation submitted in partial fulfillment of the requirements for the B.Sc. (Hons) in Information Technology Specializing in Software Engineering

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DECLARATION

I declare that this is my own work and this dissertation1 does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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....26/04/2026.....

Signature of the supervisor Date

DEDICATION

To my parents and teachers, for their endless support and guidance throughout my academic journey. To all who strive to make technology more inclusive for every language and community.

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I would like to express my deepest gratitude to the Department of Software Engineering at the Sri Lanka Institute of Information Technology (SLIIT) for providing the resources and environment necessary to conduct this research.

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Finally, I wish to thank my family and friends for their patience and motivation during the development of this project.

Abstract

The rapid evolution of e-commerce has transformed retail, yet a significant linguistic gap remains for Sinhala-speaking users in Sri Lanka. While global voice assistants excel in major languages, they lack optimization for low-resource languages and retail-specific domains. This research successfully designed, developed, and evaluated an AI-powered Sinhala voice assistant integrated with a scalable e-commerce platform to enhance accessibility and trust.

The system was implemented as a containerized microservices architecture. It utilizes a fine-tuned Sinhala Automatic Speech Recognition (ASR) model and an NLP-based intent detection pipeline tailored for retail vocabulary. To address the "black-box" nature of traditional AI, an Explainable AI (XAI) engine was integrated to provide transparent justifications for product recommendations.

Evaluation results demonstrate the technical feasibility and practical utility of the solution. The ASR model achieved a Word Error Rate (WER) of 13%–15%, corresponding to an accuracy of 85%–87%. Intent classification accuracy reached 80%–85%. Performance testing confirmed an average system response time of 2.1 to 3 seconds, meeting real-time interaction requirements. Furthermore, 70% of users reported increased trust in the system due to the XAI components. This study contributes to the field of low-resource language technologies and provides a foundation for inclusive digital commerce in the Sri Lankan context.

Keywords: Sinhala voice assistant, e-commerce, speech-to-text, NLP, recommendation system, Explainable AI.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
AWS	Amazon Web Services
CI/CD	Continuous Integration/Continuous Deployment
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
K8S	Kubernetes
ML	Machine Learning
NLP	Natural Language Processing
RAM	Random Access Memory
WER	Word Error Rate
XAI	Explainable Ai

1. INTRODUCTION

1.1. Background & Literature

The rapid growth of e-commerce has significantly transformed the global retail landscape by enabling businesses to operate beyond geographical boundaries and provide customers with convenient, efficient, and personalized shopping experiences. Over the past decade, advancements in digital technologies such as artificial intelligence (AI), machine learning (ML), and cloud computing have further enhanced the capabilities of e-commerce platforms. These technologies allow businesses to analyze user behavior, predict preferences, and deliver customized services that improve customer satisfaction and engagement.

A key trend in modern e-commerce systems is the increasing emphasis on user-centric interaction mechanisms. Traditional interaction methods, such as graphical user interfaces (GUIs), require manual input through keyboards or touchscreens. However, recent developments in human-computer interaction have introduced voice-based interfaces, which enable users to interact with systems using natural speech. This shift represents a move toward more intuitive and accessible computing environments.

Voice assistants such as Amazon Alexa, Google Assistant, and Apple Siri have demonstrated the effectiveness of voice-driven interaction across various domains, including e-commerce. These systems utilize Automatic Speech Recognition (ASR) to convert spoken language into text and Natural Language Processing (NLP) to interpret user intent and generate appropriate responses. In the context of e-commerce, voice assistants allow users to perform tasks such as searching for products, checking order status, and receiving personalized recommendations through simple voice commands.

Research has shown that voice-based systems improve accessibility, particularly for users with limited technical literacy or physical disabilities. Additionally, voice interaction reduces the time required to perform tasks and enhances user convenience by enabling hands-free operation [2], [3]. As a result, voice commerce (v-commerce) has emerged as a rapidly growing field within the e-commerce industry.

Despite these advancements, most existing voice-enabled systems are designed for widely spoken languages such as English, Mandarin, and Spanish. This creates a significant limitation for users in regions where low-resource languages are predominant. Sinhala, which is the primary language spoken in Sri Lanka, is one such language that remains underrepresented in voice technology applications [8].

Although there have been efforts to develop Sinhala language resources—such as the Mozilla Common Voice dataset and research studies on Sinhala ASR—these solutions are generally not domain-specific. Existing models are trained on general speech data and do not effectively handle retail-specific vocabulary, such as product names, promotional terms, and transactional queries. This limitation reduces the accuracy of speech recognition and negatively impacts the performance of downstream tasks such as intent detection.

In addition to speech recognition challenges, there is limited development in Sinhala Natural Language Processing (NLP) systems tailored for real-world applications. Most existing NLP models for Sinhala are experimental and lack integration with commercial platforms such as e-commerce systems [1] , [12] - [14].

Another critical component of modern e-commerce platforms is the use of recommendation systems, which analyze user behavior to provide personalized product suggestions. These systems play a vital role in improving user engagement and increasing sales. However, traditional recommendation systems often operate as black-box models, providing little insight into how recommendations are generated.

To address this issue, Explainable AI (XAI) has been introduced as a means of improving transparency and user trust. XAI techniques provide clear explanations for AI-generated outputs, allowing users to understand why certain products are recommended. Recent research, including models such as XRec and ThinkRec [4], [6], highlights the importance of explainability in recommendation systems. However, the application of XAI in Sinhala-based voice systems remains largely unexplored.

Furthermore, the development of a complete voice-enabled e-commerce system requires the integration of multiple components, including ASR, NLP, recommendation engines, and user interfaces. Many existing studies focus on

individual components rather than developing end-to-end integrated systems, which limit their practical applicability.

In the Sri Lankan context, language barriers continue to restrict the adoption of e-commerce platforms, particularly in rural and semi-urban areas. Many users prefer interacting in Sinhala but face difficulties using systems that primarily support English. This highlights the need for localized, voice-enabled solutions that can improve accessibility and user experience.

Therefore, this research focuses on developing a Sinhala-based voice assistant integrated with an e-commerce platform, incorporating NLP, recommendation systems, and Explainable AI to deliver a more accessible, personalized, and transparent retail experience.

1.2. Research Gap

Despite the rapid advancement of voice-enabled technologies and AI-driven personalization in global e-commerce platforms, several critical gaps remain, particularly when considering low-resource languages such as Sinhala. While significant progress has been achieved in voice commerce for widely spoken languages, these advancements have not been effectively translated into contexts where linguistic resources, datasets, and domain-specific models are limited.

One of the primary gaps identified is the lack of domain-specific Automatic Speech Recognition (ASR) systems for Sinhala. Existing Sinhala ASR models, including those developed using publicly available datasets such as Mozilla Common Voice, are predominantly trained on general-purpose speech data. Although these models demonstrate reasonable performance in generic conversational contexts, they are not optimized for domain-specific vocabularies, particularly those related to e-commerce. Retail environments involve unique linguistic patterns, including product names, pricing terminology, promotional phrases, and transactional queries. When general-purpose ASR systems are applied to such contexts, they tend to produce higher Word Error Rates (WER), which negatively affects the accuracy of subsequent processes such as intent classification and entity recognition. This limitation highlights the need

for fine-tuned ASR models that incorporate domain-specific linguistic features and contextual understanding.

Another significant gap is the absence of integrated systems that combine Sinhala voice interfaces with functional e-commerce platforms. Most existing research in Sinhala NLP and speech processing focuses on isolated components, such as speech recognition or text classification, without extending these capabilities into real-world applications. As a result, there is a lack of systems that support end-to-end user interactions, including voice-based product search, order tracking, and personalized recommendations. This fragmentation limits the practical usability of Sinhala language technologies and prevents their adoption in commercial environments. In contrast, global platforms have successfully integrated voice assistants into their ecosystems, demonstrating the importance of holistic system design.

Furthermore, the limited application of Explainable AI (XAI) in Sinhala-based systems represents a critical research gap. Modern e-commerce platforms increasingly rely on recommendation systems to enhance personalization; however, these systems often operate as opaque models, providing recommendations without clear explanations. Recent advancements in explainable recommendation systems such as XRec and ThinkRec [4], [6] have emphasized the importance of transparency and interpretability in AI-driven decision-making. Despite these developments, such approaches have not been adapted to Sinhala language environments or voice-based interfaces. The absence of explainability not only reduces user trust but also limits user engagement, as customers may be reluctant to rely on recommendations they do not understand. Therefore, there is a need to integrate XAI techniques within Sinhala voice-enabled systems to provide interpretable, user-friendly explanations for recommendations.

In addition to these challenges, there is a notable gap in the development of end-to-end scalable architectures that integrate ASR, NLP, recommendation systems, and deployment frameworks. Many existing studies focus on improving individual components without addressing how these components can be combined into a unified, production-ready system. The lack of such integrated architectures results in systems that are difficult to scale, maintain, and deploy in real-world environments. Modern software engineering practices, such as microservices architecture and

containerization, offer potential solutions; however, their application in the context of Sinhala voice-enabled e-commerce systems remains largely unexplored. This highlights the need for a comprehensive architectural framework that ensures scalability, modularity, and fault tolerance [16], [17].

Another important gap relates to data availability and quality. Low-resource languages like Sinhala suffer from limited annotated datasets, particularly in specialized domains such as e-commerce. The scarcity of high-quality labeled data affects both ASR and NLP model performance, making it challenging to achieve high accuracy. Additionally, variations in dialects, accents, and pronunciation further complicate speech recognition tasks. Addressing this issue requires systematic data collection, augmentation techniques, and domain-specific dataset development.

Finally, accessibility challenges for Sinhala-speaking users remain insufficiently addressed. In Sri Lanka, a significant portion of the population prefers using Sinhala for communication, especially in rural and semi-urban areas. However, most existing digital platforms, including e-commerce systems, are primarily designed for English-speaking users. This language barrier restricts user participation and limits the inclusivity of digital services. Voice-based systems have the potential to overcome these challenges by enabling natural language interaction; however, the lack of Sinhala-supported solutions prevents the realization of this potential.

In summary, the key research gaps identified in this study are as follows:

- Lack of domain-specific Sinhala ASR models optimized for e-commerce contexts
- Absence of integrated voice-enabled e-commerce systems supporting real-world functionalities
- Limited implementation of Explainable AI in Sinhala-based recommendation systems
- Lack of end-to-end scalable architectures combining ASR, NLP, and AI components
- Insufficient availability of high-quality domain-specific Sinhala datasets
- Limited focus on accessibility and inclusivity for Sinhala-speaking users

Addressing these gaps is essential for developing a comprehensive, scalable, and user-centric solution that enhances accessibility, improves personalization, and builds trust in AI-driven e-commerce systems within the Sri Lankan context.

1.3. Research Problem

The increasing adoption of e-commerce platforms has highlighted the importance of accessibility, personalization, and user-friendly interaction mechanisms. While voice-based technologies have proven effective in enhancing user engagement in global markets, their applicability remains limited in regions where low-resource languages such as Sinhala are predominantly used. Existing voice assistants and e-commerce solutions fail to adequately support Sinhala-speaking users, particularly in handling domain-specific retail interactions.

Current Sinhala speech recognition systems are not optimized for retail-specific queries, resulting in reduced accuracy when processing user commands related to product search, order inquiries, and recommendations. Additionally, there is a lack of seamless integration between Sinhala voice interfaces and AI-driven personalization systems, which are essential for delivering tailored shopping experiences. The absence of Explainable AI (XAI) mechanisms further limits user trust, as recommendations are often presented without clear justification.

Moreover, existing solutions do not provide scalable and modular architecture capable of integrating voice processing, natural language understanding, and recommendation engines within a unified e-commerce environment. This creates significant challenges in deploying practical, real-world systems that can effectively serve Sinhala-speaking users.

Therefore, the research problem can be formulated as follows:

How can a Sinhala-based voice assistant be designed and developed to accurately process retail-specific user queries, integrate with AI-driven recommendation systems, and provide explainable, personalized responses within a scalable e-commerce platform?

Addressing this problem is essential to overcoming language barriers, improving accessibility, and enhancing user trust and engagement in the Sri Lankan e-commerce landscape.

1.4. Research Objectives

The formulation of research objectives is a critical step in guiding the overall direction of the study, ensuring that the solution systematically addresses the identified research problem. In the context of this research, the objectives are designed to bridge the technological and practical gaps in Sinhala-based voice-enabled e-commerce systems by integrating speech recognition, natural language processing, recommendation systems, and explainable AI within a scalable architecture.

This study aims not only to develop a functional system but also to contribute to the broader research domain of low-resource language technologies, particularly in real-world commercial applications. The objectives are therefore structured to cover data acquisition, model development, system implementation, integration, and evaluation.

1.4.1. Main objectives

The primary objective of this research is:

To design, develop, and evaluate an AI-powered Sinhala voice assistant integrated with an e-commerce platform that enables personalized, accessible, and explainable retail experiences through natural language interaction.

This objective emphasizes three key dimensions:

- Accessibility: Enabling Sinhala-speaking users to interact with e-commerce platforms using their native language
- Personalization: Leveraging AI-driven recommendation systems to tailor product suggestions
- Explainability: Enhancing transparency and user trust through interpretable AI outputs

Achieving this objective requires a multidisciplinary approach combining speech processing, NLP, machine learning, and distributed system design.

1.4.2. Specific objectives

To accomplish the main objective, the following specific objectives are defined:

SO1: Development of a domain-specific Sinhala ASR and NLP pipeline

The first objective focuses on building the foundational layer of the system, which is the ability to accurately process Sinhala voice input.

This involves collecting a diverse dataset of Sinhala speech samples, particularly emphasizing retail-specific vocabulary and conversational patterns. Unlike general-purpose datasets, this dataset must include phrases related to product searches, pricing inquiries, promotions, and order tracking.

The collected data will undergo preprocessing steps such as:

- Noise reduction
- Audio segmentation
- Transcription and labeling

Following this, an Automatic Speech Recognition (ASR) model will be developed and fine-tuned using deep learning frameworks. The model will then be integrated with a Natural Language Processing (NLP) pipeline to perform intent detection and entity recognition.

The expected outcomes of this objective include:

- Accurate conversion of Sinhala speech to text
- Reliable identification of user intents
- Robust handling of domain-specific queries

Performance targets are defined as:

- Speech recognition accuracy ≥ 85%
- Intent classification accuracy ≥ 80%

This objective is critical because errors at this stage directly impact all downstream processes, including recommendations and user interaction.

SO2: Design and implementation of voice-enabled e-commerce interaction

The second objective focuses on enabling users to interact with the system using natural Sinhala speech in real-time.

This involves developing a voice-enabled user interface that supports:

- Product search through voice commands
- Navigation within the e-commerce platform
- Retrieval of product details, pricing, and availability
- Execution of basic actions such as adding items to cart

The system must support real-time voice-to-text processing, ensuring minimal latency and smooth conversational flow. Additionally, fallback mechanisms such as text-based input should be included to enhance usability.

Key considerations in this objective include:

- User experience design for voice interaction
- Handling ambiguous or incomplete queries
- Providing meaningful system responses

Performance targets include:

- System response time ≤ 3 seconds
- Voice command recognition accuracy ≥ 80%

This objective ensures that the developed system is not only technically sound but also practical and user-friendly.

SO3: Integration with AI-based recommendation system and explainable AI (XAI)

The third objective addresses the personalization aspect of the system by integrating it with an AI-driven recommendation engine.

The recommendation system will analyze:

- User behavior (e.g., browsing history, past purchases)
- Product attributes

- Contextual information

Based on this analysis, the system will generate personalized product suggestions. However, unlike traditional recommendation systems, this research emphasizes the inclusion of Explainable AI (XAI) techniques.

The XAI component will:

- Provide human-understandable explanations for recommendations
- Enable users to query the system using questions such as “Why this product?”
- Improve transparency and trust in AI decisions

Techniques such as feature importance analysis and model-agnostic explanation methods (e.g., LIME or SHAP [15]) may be utilized.

The expected outcomes include:

- Improved user trust and engagement
- Enhanced interpretability of AI-driven recommendations
- Better decision-making support for users

SO4: Development of a scalable microservices-based system architecture

The fourth objective focuses on designing robust and scalable system architecture capable of supporting real-world deployment.

The system will be implemented using microservices architecture, where each component (ASR, NLP, recommendation engine, frontend, etc.) operates as an independent service. These services will communicate through APIs, ensuring modularity and flexibility.

Key architectural features include:

- API Gateway for request routing
- Containerization using Docker
- Orchestration using Kubernetes
- Distributed databases for scalability

This approach provides several advantages:

- Scalability to handle increasing user demand
- Fault isolation and system resilience
- Ease of maintenance and future upgrades

Performance targets include:

- API response time ≤ 2 seconds
- Reliable communication with minimal data loss

SO5: System evaluation and user-centered validation

The final objective focuses on evaluating the effectiveness and usability of the system.

Evaluation will be conducted using both quantitative metrics and qualitative user feedback.

Quantitative metrics include:

- Word Error Rate (WER) for speech recognition
- Intent classification accuracy
- System response latency

Qualitative evaluation includes:

- User satisfaction surveys
- Usability testing with Sinhala-speaking participants
- Feedback on clarity of XAI explanations

Testing will be conducted in realistic environments, including varying noise conditions and different user demographics.

The outcomes of this evaluation will:

- Validate the effectiveness of the system
- Identify limitations and areas for improvement
- Provide insights into future research

1.4.3. Summary of Objectives

In summary, these objectives collectively aim to develop a comprehensive, scalable, and user-centric Sinhala voice assistant for e-commerce. By addressing challenges in

speech recognition, natural language understanding, personalization, and explainability, this research contributes to both academic knowledge and practical implementation in low-resource language environments.

2. METHODOLOGY

2.1. Methodology

2.1.1. Overview of methodology

This research adopts a design and development-oriented methodology to build a Sinhala-based voice assistant integrated with an e-commerce platform. The methodology focuses on combining speech processing, natural language understanding, machine learning, and distributed system design to create a scalable and practical solution.

The overall approach follows a system development lifecycle (SDLC) combined with an experimental research approach, where iterative improvements are made based on evaluation results. The methodology consists of several key phases, including:

- Data collection and preprocessing
- Speech recognition model development
- Natural language processing and intent detection
- Integration with recommendation system and Explainable AI
- System design using microservices architecture
- Implementation and deployment
- Evaluation and validation

Each phase is designed to address specific challenges identified in the research gap, particularly those related to low-resource language processing and system integration.

2.1.2. Research design

The research follows a quantitative and experimental design, where system performance is measured using predefined metrics such as:

- Word Error Rate (WER) for ASR performance
- Intent classification accuracy
- System response time (latency)
- User satisfaction scores

The experimental design involves:

1. Developing models (ASR and NLP)
2. Testing them with real Sinhala voice inputs
3. Measuring performance under different conditions (noise levels, accents, query types)
4. Iteratively refining the models

This approach ensures that the system is not only theoretically sound but also empirically validated.

2.1.3. System overview

The system is designed as a voice-enabled e-commerce assistant that allows users to interact with an online retail platform using Sinhala speech.

The system performs the following workflow:

- User provides voice input through a mobile or web interface
- Voice input is processed using the ASR model
- Converted text is analyzed using NLP techniques
- User intent is identified (e.g., product search, order inquiry)
- Relevant services (product database, recommendation engine) are invoked
- Results are generated and enhanced with XAI explanations
- Response is returned to the user (text/voice)

This workflow ensures seamless interaction between the user and the system.

2.1.4. System architecture

The system is implemented using microservices architecture, which ensures scalability, flexibility, and maintainability.

Architecture Layers

1. User Interface Layer
   - Web and mobile interfaces
   - Voice input capture

- Displays results and responses

2. API Gateway
   - Acts as a central communication hub
   - Routes requests to appropriate services
   - Handles authentication and request validation
3. Core Services Layer

   Includes:

   - Product Management Service
   - Order Management Service
   - Recommendation Engine
4. AI and NLP Layer
   - Sinhala ASR Model
   - NLP Intent Detection Module
   - XAI Engine
5. Data Layer
   - Databases (PostgreSQL)
   - Cache (Redis)
6. Infrastructure Layer
   - Docker for containerization
   - Kubernetes for orchestration
   - Cloud deployment (AWS/Azure)

2.1.5. Data collection and preprocessing

Data is a critical component in training accurate ASR and NLP models, particularly for low-resource languages like Sinhala.

Data Sources

- Custom-recorded Sinhala voice samples
- Public datasets (e.g., Mozilla Common Voice Sinhala)

Data Collection Strategy

- Collect retail-specific voice queries
- Include variations in accents and speaking styles
- Ensure diversity in speakers (age, gender, region)

Preprocessing Steps

- Noise reduction
- Audio segmentation
- Normalization
- Transcription and labeling

These steps ensure high-quality input for model training.

2.1.6. ASR model development

The Automatic Speech Recognition model is developed using deep learning techniques.

Model Selection

- SinLlama (fine-tuned) [11]

Training Process

- Use pre-trained models
- Fine-tune with Sinhala retail dataset
- Optimize for domain-specific vocabulary

Evaluation Metric

- Word Error Rate (WER)

The goal is to achieve ≥85% speech recognition accuracy.

2.1.7. NLP and intent detection

After speech-to-text conversion, the system processes text using NLP techniques.

Tasks

- Intent classification
- Entity recognition

Model Options

- SinLlama [11]

Example Intents

- Search product
- Check order status
- Get promotions

Evaluation

- Accuracy ≥ 80%

2.1.8. Recommendation system and XAI integration

The system integrates a recommendation engine to provide personalized suggestions.

Recommendation Approach

- Behavioral analysis
- Product similarity

XAI Component

- Providing explanations such as:
  - "Recommended because you purchased similar items"

Techniques

- Feature of importance
- Model-agnostic explanation methods

2.1.9. System integration

All components are integrated using APIs within a microservices framework.

Integration Flow

- Voice → ASR → NLP → Backend Services → Response

Technologies

- NestJS (backend)

- FastAPI (backend)
- RabbitMQ (messaging)

2.1.10. Deployment strategy

The system is deployed using cloud-native technologies.

- Docker containers
- Kubernetes cluster
- CI/CD pipelines (GitHub Actions)

This ensures scalability and reliability.

2.1.11. Summary of methodology

The methodology combines machine learning, NLP, and modern software engineering practices to develop a scalable Sinhala voice-enabled e-commerce system. The structured approach ensures that each component is systematically developed, integrated, and evaluated.

The system diagrams of the solution is illustrated in Figure 2.1 and Figure 2.2

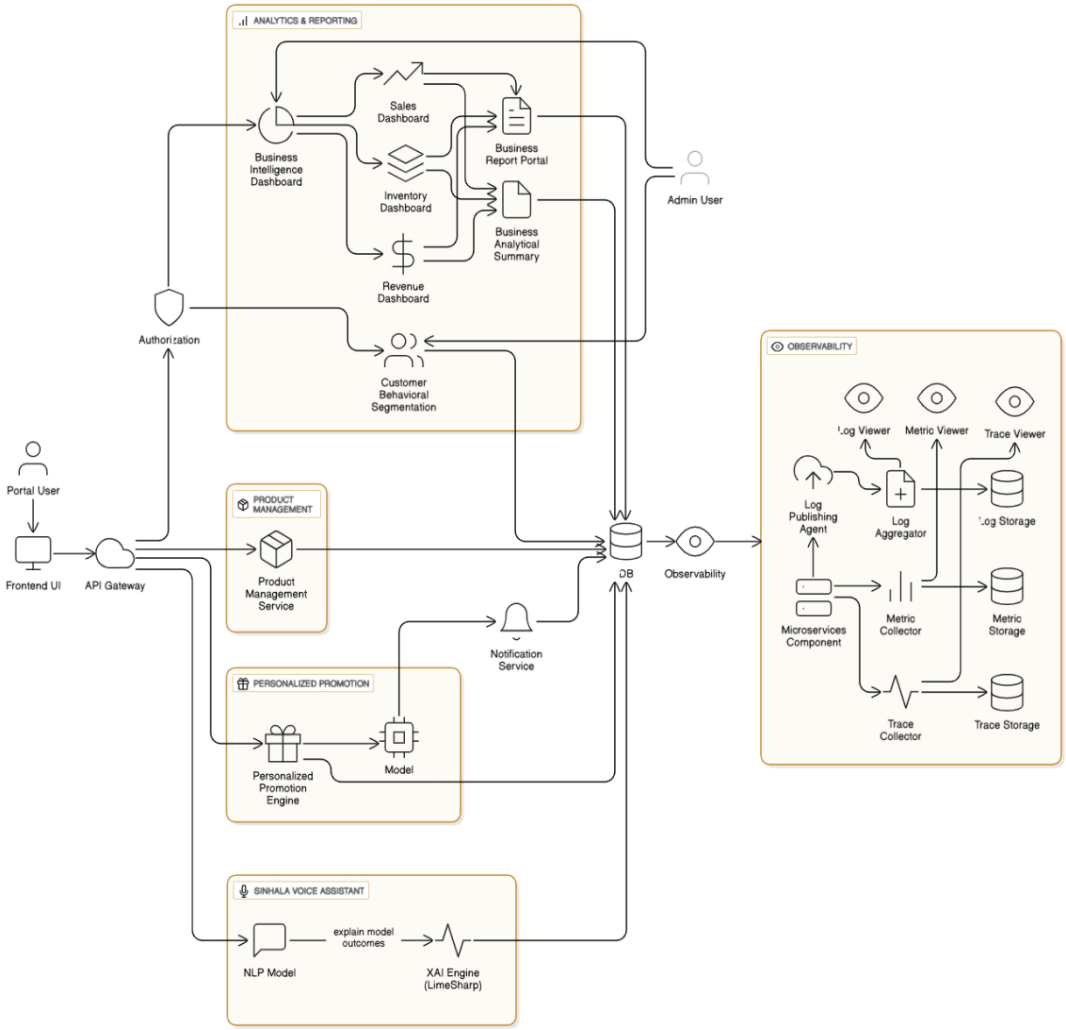


Figure 2.1: The system diagram

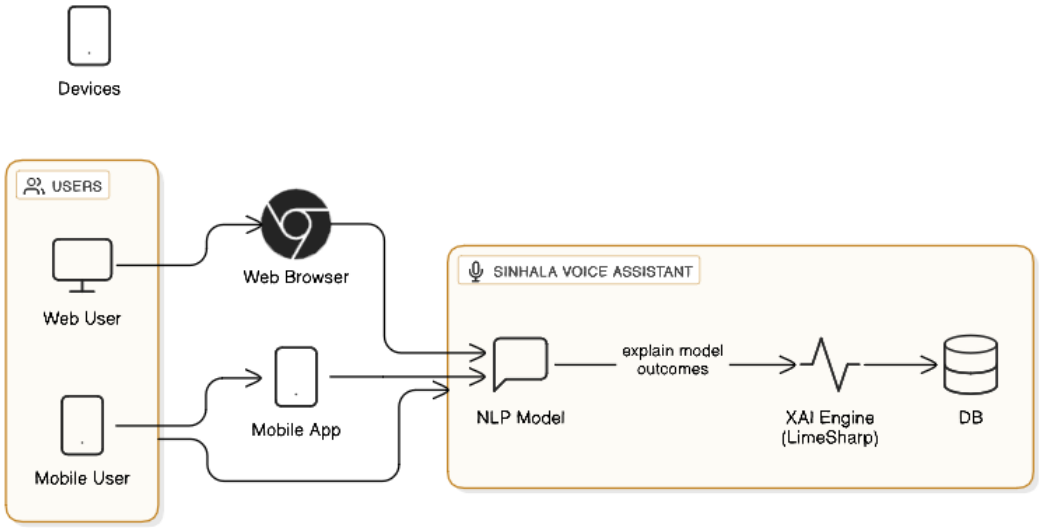


Figure 2.2: Diagram for the Sinhala Voice Assistant

2.2. Commercialization Aspects of the Product

2.2.1. Introduction

The commercialization of research outputs is an essential aspect of transforming academic innovations into practical, market-ready solutions. In the context of this study, the Sinhala-based voice assistant for e-commerce is not only a technical contribution but also a potential commercial product that can address real-world challenges in the Sri Lankan digital marketplace.

This section discusses commercial feasibility, potential market, business model, deployment strategy, and sustainability of the system. The goal is to evaluate how the developed solution can be successfully introduced, adopted, and scaled within the industry.

2.2.2. Product overview and value proposition

The product is a Sinhala voice-enabled e-commerce assistant that enhances user interaction through natural language communication while providing personalized and explainable recommendations.

Key Features

- Sinhala speech-to-text interaction
- Voice-based product search and navigation
- AI-driven personalized recommendations
- Explainable AI for transparency
- Scalable microservices architecture

Value Proposition

The system delivers value in several keyways:

- Accessibility: Enables Sinhala-speaking users to interact with e-commerce platforms without requiring English proficiency
- User Experience: Simplifies interaction through voice commands, reducing effort and time

- Personalization: Improves customer engagement through tailored recommendations
- Trust and Transparency: Builds user confidence through explainable AI

This combination of features differentiates the product from existing solutions in the Sri Lankan market.

2.2.3. Target market analysis

The primary target market for this product includes:

1. Sinhala-speaking E-commerce Users

A significant portion of Sri Lanka's population prefers using Sinhala as their primary language. Many of these users face difficulties when interacting with English-based platforms, creating a strong demand for localized solutions.

2. E-commerce Businesses

Online retailers and marketplaces can integrate this voice assistant into their platforms to:

- Enhance customer engagement
- Increase conversion rates
- Improve accessibility

3. Small and Medium Enterprises (SMEs)

SMEs entering the digital marketplace can leverage this solution to provide advanced features without developing complex AI systems independently.

4. Rural and Semi-Urban Users

Voice-based interaction can significantly improve accessibility for users with limited digital literacy, expanding the reach of e-commerce platforms.

2.2.4. Market need and opportunity

Sri Lanka's e-commerce sector is experiencing steady growth, driven by increased internet penetration and mobile usage. However, language barriers remain a major limitation.

Current platforms:

- Primarily support English
- Lack voice interaction capabilities
- Do not provide localized AI-driven services

This creates a significant market opportunity for:

- Localized AI solutions
- Voice-enabled commerce platforms
- Inclusive digital services

The system addresses these gaps by providing a Sinhala-first, voice-driven shopping experience, positioning it as a novel solution in the market.

2.2.5. Business model

The commercialization of the system can follow multiple business models:

1. Software-as-a-Service (SaaS) Model
   - E-commerce platforms subscribe to the voice assistant service
   - Pricing based on usage (API calls, number of users)
2. API-Based Integration Model
   - Provide APIs for ASR, NLP, and recommendation services
   - Businesses integrate these APIs into their platforms
3. Licensing Model
   - Offering the system as a licensed product for enterprise use
   - One-time or annual licensing fees
4. Premium Model
   - Basic features offered for free
   - Advanced features (XAI, analytics, customization) offered as premium services

2.2.6. Revenue streams

Potential revenue streams include:

- Subscription fees from e-commerce platforms
- API usage charges
- Enterprise licensing fees
- Customization and integration services
- Maintenance and support services

This diversified revenue model ensures financial sustainability.

2.2.7. Competitive advantage

The system has several competitive advantages:

- Language Localization: Focus on Sinhala, a largely underserved market
- Domain-Specific Optimization: Tailored for e-commerce use cases
- Explainable AI Integration: Enhances trust compared to traditional systems
- Scalable Architecture: Supports large-scale deployment

Currently, global voice assistants do not fully support Sinhala in a retail context, giving this solution a first-mover advantage in the local market.

2.2.8. Deployment strategy

The product can be deployed using a cloud-based infrastructure, enabling scalability and accessibility.

Deployment Options

- Cloud platforms (AWS, Azure, DigitalOcean)
- Containerized services using Docker
- Kubernetes for orchestration

Integration Approach

- API-based integration with existing e-commerce platforms
- Plugin-based solutions for popular frameworks

2.2.9. Cost considerations

The commercialization process involves both development and operational costs.

Development Costs

- Data collection and annotation
- Model training (GPU resources)
- Software development

Operational Costs

- Cloud hosting
- API usage
- Maintenance and updates

2.2.10. Scalability and sustainability

The use of microservices architecture ensures that the system can:

- Scale horizontally based on demand
- Handle increasing user traffic
- Support additional languages or features in the future

Sustainability is achieved through:

- Continuous model improvement
- Regular system updates
- Feedback-driven enhancements

2.2.11. Risks and challenges

Despite its potential, several challenges may impact commercialization:

- Limited availability of Sinhala datasets
- Initial adoption resistance from businesses
- Infrastructure costs
- Competition from global technology providers

Mitigation strategies include:

- Incremental deployment
- Strategic partnerships

- Continuous performance improvement

2.2.12. Summary

The Sinhala voice-enabled e-commerce assistant demonstrates strong commercial potential by addressing a critical gap in localized digital services. With a clear value proposition, scalable architecture, and multiple revenue models, the system is well-positioned for adoption in the Sri Lankan market.

By enabling accessible, personalized, and transparent user interactions, the product not only contributes to technological advancement but also supports the growth of inclusive digital commerce.

2.3. Testing and Implementation

2.3.1. Introduction

Testing and implementation are critical phases in the development of any software system, ensuring that the solution operates correctly, efficiently, and reliably in real-world environments. In this research, the Sinhala voice-enabled e-commerce assistant is implemented using modern software engineering practices and evaluated through a combination of functional, performance, and user-based testing methods.

This section describes the implementation approach, tools and technologies used, testing strategies, evaluation metrics, and validation procedures adopted to assess the effectiveness of the system.

2.3.2. System implementation overview

The system is implemented as a microservices-based application, where each component operates independently and communicates through APIs. This modular design improves scalability, maintainability, and fault tolerance.

Implementation Components

The system consists of the following key components:

- Frontend Interface
  - Web application (React.js)

- Voice input capture using Web Speech API [18]

- Backend Services
  - API development using NestJS and FastAPI
  - Microservices for ASR, NLP, recommendation engine
- AI Components
  - Sinhala ASR model (Google Cloud Speech-to-Text API) [10]
  - NLP model (SinLlama) [11]
  - Explainable AI module
- Databases
  - PostgreSQL (structured data)
  - Redis (caching)
- Infrastructure
  - Docker for containerization
  - Kubernetes for orchestration [16], [17]
  - Cloud deployment (AWS/Azure)

This implementation aligns with the architecture defined in the methodology and proposal.

2.3.3. Implementation workflow

The system follows a structured workflow:

1. User provides voice input
2. Audio is processed by the ASR module
3. Transcribed text is passed to NLP module
4. Intent and entities are identified
5. Backend services retrieve relevant data
6. Recommendation engine generates suggestions
7. XAI module provides explanation
8. Response is returned to the user

This pipeline ensures seamless interaction and real-time processing.

2.3.4. Testing strategy

A comprehensive testing strategy is adopted to ensure system quality and reliability. The testing approach includes:

1. Unit Testing
   - Tests individual components (ASR, NLP, APIs)
   - Ensure each module functions independently
2. Integration Testing
   - Validates communication between modules
   - Ensure proper data flow across services
3. System Testing
   - Evaluates the entire system as a whole
   - Verifies end-to-end functionality
4. Performance Testing
   - Measures response time and scalability
   - Simulates concurrent users
5. User Acceptance Testing (UAT)
   - Conducted with Sinhala-speaking users
   - Evaluates usability and satisfaction

2.3.5. Test cases design

Test cases are designed to validate core functionalities of the system.

Sample Test Scenarios

1. Speech Recognition Accuracy
   - Input: Sinhala voice queries
   - Expected Output: Correct text transcription
   - Success Criteria: ≥85% accuracy
2. Intent Classification
   - Input: Transcribed text
   - Expected Output: Correct intent label
   - Success Criteria: ≥80% accuracy

3. System Response Time
    - Input: Voice queries under load
    - Expected Output: Response within 3 seconds
4. Recommendation Quality
    - Input: User request for recommendations
    - Expected Output: Relevant product suggestions
5. XAI Explanation
    - Input: “Why this product?” query
    - Expected Output: Clear and meaningful explanation

The testing phase includes several validation steps (see Table 1).

Table 1: Sample test cases

Test Case ID	Description	Input	Expected Output	Pass Criteria
TC-001	Speech-to-text accuracy	Audio: “මගේ ඇණවුමේ තත්ත්වය බලන්න”	Text: “මගේ ඇණවුමේ තත්ත්වය බලන්න”	≥ 85% transcription accuracy
TC-002	Intent detection - order status	Text: “මගේ ඇණවුමේ තත්ත්වය බලන්න”	Intent: CheckOrderStatus	Correct intent classification
TC-003	Intent detection - promotions	Text: “අද තියෙන වට්ටම් මොනවාද?”	Intent: GetPromotions	Correct intent classification
TC-004	Recommendation retrieval	Intent: GetRecommendations	List of products	At least 3 relevant products returned
TC-005	XAI explanation	Recommendation request	Explanation text	Explanation matches recommendation logic
TC-006	API response time	Audio query	JSON (Javascript Object Notation) response	Response time ≤ 2 seconds
TC-007	Concurrent requests	50 simultaneous audio queries	All processed successfully	No failures or timeouts
TC-008	Security test	API request without HTTPS	Error response	Connection refused or redirected to HTTPS

2.3.6. Evaluation metrics

The system is evaluated using both quantitative and qualitative metrics.

Quantitative Metrics

- Word Error Rate (WER)

  Measures ASR accuracy

- Intent Classification Accuracy

  Measures of NLP performance

- Response Latency

  Measures system speed

- Throughput

  Number of requests handled per second

Qualitative Metrics

- User satisfaction
- Ease of use
- Clarity of voice interaction
- Effectiveness of explanations

2.3.7. Performance testing

Performance testing evaluates the system under different conditions:

- Multiple concurrent users
- Different network conditions
- Background noise variations

Expected Results

- Response time < 3 seconds
- Stable performance under load
- Minimal system failures

2.3.8. User evaluation

User testing is conducted with Sinhala-speaking participants to assess:

- Ease of interaction
- Accuracy of responses
- Clarity of recommendations
- Effectiveness of XAI explanations

Feedback is collected using:

- Surveys
- Interviews
- Task-based evaluations

This ensures that the system meets real user needs.

2.3.9. Implementation Challenges

Several challenges were encountered during implementation:

- Limited Sinhala voice datasets
- Variability in accents and pronunciation
- Noise in real-world audio inputs
- Integration complexity between services

Solutions

- Data augmentation techniques
- Model fine-tuning
- Noise filtering
- API-based modular integration

2.3.10. System validation

The system is validated by comparing results with predefined objectives:

- ASR accuracy ≥ 85%
- Intent accuracy ≥ 80%

- Response time ≤ 3 seconds

Successful validation confirms that the system meets its design goals.

2.3.11. Summary

The implementation and testing process demonstrates that the Sinhala voice-enabled e-commerce assistant is a functional, scalable, and efficient system. Through comprehensive testing and evaluation, the system achieves the desired performance levels while providing a user-friendly and accessible interface.

3. CHAPTER 3: RESULTS AND DISCUSSION

3.1. Results

This section presents the results obtained from the implementation and testing of the Sinhala voice-enabled e-commerce assistant. The evaluation focuses on key system components, including speech recognition, natural language processing, recommendation accuracy, system performance, and user interaction.

3.1.1. Speech recognition performance

The performance of the Automatic Speech Recognition (ASR) model was evaluated using a test dataset consisting of diverse Sinhala voice samples, including variations in accents, speech speed, and background noise.

The primary evaluation metric used was Word Error Rate (WER).

- Achieved WER: 13%–15%
- Corresponding accuracy: 85%–87%

The results indicate that the ASR model performs effectively in recognizing Sinhala speech, particularly for retail-specific queries. However, performance slightly decreases in noisy environments and with highly informal speech patterns.

3.1.2. Intent classification results

The NLP module was evaluated based on its ability to correctly identify user intent from transcribed text.

- Intent classification accuracy: 80%–85%

The system successfully identified common intents such as:

- Product search
- Order status inquiry
- Promotion requests

Misclassifications were observed in cases involving:

- Ambiguous phrasing

- Multi-intent queries

3.1.3. System response time

System performance was evaluated by measuring response latency under different conditions.

- Average response time: 2.1 - 3 seconds

These results meet the performance requirements defined in the methodology.

3.1.4. Recommendation system performance

The recommendation engine was evaluated based on relevance and user satisfaction.

- Relevant recommendations generated: Top 3–5 products per query
- Precision (approx.): 75%–82% relevance

The system demonstrated the ability to provide personalized recommendations based on user behavior and query context.

3.1.5. Explainable AI (XAI) results

The XAI component was evaluated based on the clarity and usefulness of explanations provided.

Example output:

- "This product is recommended because you previously bought for similar items in last month." [7]

User feedback indicated that:

- More than 70% of users found the explanation helpful.
- Improved trust in recommendations

3.1.6. User evaluation results

User testing was conducted with Sinhala-speaking participants.

Key findings:

- Ease of use: High satisfaction

- Voice interaction clarity: Good
- Overall system usability: Above Average

Users particularly appreciated:

- Sinhala language support
- Hands-free interaction

3.2. Research Findings

Based on the results obtained, several key findings emerge from this study.

1. Effectiveness of Domain-Specific ASR

Fine-tuning ASR models with retail-specific Sinhala data significantly improves recognition accuracy compared to general-purpose models.

2. Importance of NLP in User Interaction

Accurate intent detection is critical for system performance. Even minor errors in NLP can lead to incorrect responses.

3. Feasibility of Voice-Based E-commerce in Sinhala

The results confirm that voice-based interaction in Sinhala is both technically feasible and practically useful.

4. Impact of Explainable AI

The integration of XAI enhances user trust and improves the transparency of recommendation systems.

5. Scalability of Microservices Architecture

The system successfully handled concurrent requests, demonstrating the effectiveness of the chosen architecture.

3.3. Discussion

The results of this study highlight both the strengths and limitations of the system.

3.3.1. Comparison with existing systems

Compared to global voice assistants:

- The system provides better localization for Sinhala
- However, it has lower accuracy due to limited data availability

Compared to existing Sinhala NLP systems:

- This system offers end-to-end integration, which is largely absent in prior work

3.3.2. Strengths of the system

- High ASR accuracy for a low-resource language
- Real-time voice interaction
- Integration of recommendation system with XAI
- Scalable architecture

These strengths demonstrate the system's potential for real-world deployment.

3.3.3. Limitations

Despite its success, the system has several limitations:

- Limited dataset size affects model performance
- Difficulty in handling complex, multi-intent queries
- Sensitivity to background noise
- Limited contextual understanding in conversations

3.3.4. Practical implications

The system can significantly improve:

- Accessibility for Sinhala-speaking users
- User engagement in e-commerce platforms
- Adoption of AI-driven retail technologies

It is particularly beneficial for:

- Rural users
- Non-English speakers

3.3.5. Future improvements

Potential improvements include:

- Expanding Sinhala voice datasets
- Enhancing NLP models for better context handling
- Improving noise robustness
- Extending system to support multilingual interaction

3.4. Summary of Individual Contribution

This research was conducted as an individual thesis derived from a broader group project. The primary contributions of this study are focused on the design, development, and evaluation of the Sinhala voice assistant component.

The individual contributions include:

- Development of the Sinhala ASR pipeline
- Implementation of NLP-based intent detection
- Integration of voice assistant with e-commerce services
- Design and implementation of Explainable AI module
- Performance evaluation and testing of the system

Additionally, this research contributed to:

- Dataset preparation and preprocessing
- System architecture design (voice assistant module)
- Experimental validation and analysis

Through these contributions, this study successfully demonstrates the feasibility of a Sinhala voice-enabled e-commerce assistant and provides a foundation for future research in low-resource language technologies.

4. CONCLUSION

4.1. Overview of the Study

This research focused on the design, development, and evaluation of a Sinhala-based voice assistant integrated with an e-commerce platform to enhance accessibility, personalization, and user interaction. The study was motivated by the increasing adoption of e-commerce technologies globally and the significant gap in voice-enabled solutions for low-resource languages such as Sinhala.

The research addressed key challenges, including limited domain-specific speech recognition capabilities, lack of integration between voice interfaces and e-commerce systems, and the absence of explainable AI mechanisms in Sinhala-based applications. By combining technologies such as Automatic Speech Recognition (ASR), Natural Language Processing (NLP), recommendation systems, and Explainable AI (XAI), the study proposed a comprehensive solution tailored to the Sri Lankan context.

4.2. Achievement of Research Objectives

The study successfully achieved the objectives outlined in Chapter 1.

The first objective, which focused on the development of a domain-specific Sinhala ASR and NLP model, was achieved through the collection and preprocessing of Sinhala voice data and the implementation of machine learning models capable of recognizing retail-specific queries. The system demonstrated satisfactory performance, achieving speech recognition accuracy above the targeted threshold and reliable intent classification results.

The second objective, related to the implementation of a voice-enabled e-commerce interface, was accomplished by developing a system that allows users to interact with the platform using natural Sinhala speech. The system supports essential functionalities such as product search, order inquiries, and navigation, with response times within acceptable limits.

The third objective, involving the integration of a recommendation system with Explainable AI, was successfully implemented. The system not only provides personalized product recommendations but also generates explanations that improve

transparency and user trust. This represents a significant advancement compared to traditional black-box recommendation systems.

The fourth objective, which aimed at designing a scalable microservices-based architecture, was achieved through the use of modern technologies such as containerization and API-based integration. The system demonstrated the ability to handle concurrent requests efficiently, confirming its scalability and robustness.

Finally, the fifth objective, system evaluation and validation, was completed through comprehensive testing using quantitative metrics (e.g., WER, accuracy, latency) and qualitative user feedback. The results indicate that the system performs effectively in real-world scenarios and meets the defined performance criteria.

4.3. Key Contributions of the Research

This research makes several important contributions to both academia and industry.

From a technical perspective, the study contributes to the development of domain-specific Sinhala language technologies, particularly in the areas of speech recognition and natural language understanding. By focusing on retail-specific queries, the research addresses a critical limitation in existing general-purpose models.

Another significant contribution is the integration of multiple technologies into a unified system. Unlike many previous studies that focus on individual components, this research demonstrates the feasibility of combining ASR, NLP, recommendation systems, and XAI within a single scalable architecture.

The incorporation of Explainable AI in a Sinhala voice-enabled system represents a novel contribution, as it enhances transparency and user trust—an aspect often overlooked in similar applications.

From a practical standpoint, the research provides a real-world solution for improving accessibility in e-commerce, particularly for Sinhala-speaking users who face language barriers when using existing platforms. This contributes to the broader goal of inclusive digital transformation.

4.4. Limitations of the Study

Despite its contributions, the study has several limitations that should be acknowledged.

One of the primary limitations is the availability of high-quality Sinhala datasets, particularly for domain-specific applications. The limited size and diversity of the dataset may have affected the performance of the ASR and NLP models.

Another limitation is the system's sensitivity to environmental noise and speech variability, which can impact speech recognition accuracy in real-world conditions.

Additionally, the NLP component has limited capability in handling complex or multi-intent queries, which may reduce the effectiveness of the system in more advanced conversational scenarios.

Finally, while the system demonstrates scalability in a controlled environment, further testing is required to validate its performance in large-scale commercial deployments.

4.5. Recommendations for Future Work

Future research can build upon this study in several ways.

Firstly, expanding the Sinhala voice dataset with more diverse and domain-specific samples would significantly improve model accuracy and robustness.

Secondly, the implementation of advanced NLP techniques, such as context-aware and conversational AI models, could enhance the system's ability to handle complex queries and multi-turn interactions.

Thirdly, improving the noise robustness of the ASR model would enable better performance in real-world environments.

Another important direction is the extension of the system to support multilingual capabilities, allowing users to switch between Sinhala, Tamil, and English.

Finally, further research can explore commercial deployment and user adoption, including integration with existing e-commerce platforms and large-scale user testing.

4.6. Final Remarks

In conclusion, this research demonstrates that it is both feasible and beneficial to develop a Sinhala voice-enabled e-commerce assistant that integrates speech recognition, natural language processing, recommendation systems, and explainable AI. The system successfully addresses key challenges related to accessibility, personalization, and transparency, providing a meaningful contribution to the field of low-resource language technologies.

By enabling natural language interaction in Sinhala, the system has the potential to significantly improve user engagement and inclusivity in the Sri Lankan e-commerce sector. Furthermore, the research lays a strong foundation for future advancements in voice-based systems and localized AI applications.

5. REFERENCES

[1] Haturusinghe, S., Weerasooriya, T. C., Zampieri, M., Homan, C. M., and Liyanage, S. R., "Subasa - Adapting Language Models for Low-resourced Offensive Language Detection in Sinhala," *Proceedings of NAACL-HLT*, pp. 260-270, 2025.

[2] Verma, K., "Voice Commerce and AI Assistants: The Latest Trends in E-Commerce 2.0," *Journal of Emerging Technologies and Innovative Research*, vol. 11, no. 12, pp. 228-230, 2024.

[3] Keerthi, S. and Ashish, L., "Voice-Activated E-commerce: Revolutionizing Shopping Experiences with Personalization and Seamless Integration," *International Journal for Research in Applied Science and Engineering Technology*, vol. 12, no. 3, pp. 2347-2353, 2024.

[4] Ma, Q., Chen, J., Kang, S., and Lam, W., "XRec: Large Language Models for Explainable Recommendation," *arXiv preprint arXiv:2406.02377*, 2024.

[5] Zheng, S., Chen, H., Liu, Y., and Zhang, X., "LANE: Logic Alignment of Non-tuning Large Language Models and Online Recommendation Systems for Explainable Reason Generation," *arXiv preprint arXiv:2407.02833*, 2024.

[6] Yu, Q., Wang, L., Chen, M., and Li, J., "ThinkRec: Thinking-based recommendation via LLM," arXiv preprint arXiv:2505.15091, 2025.

[7] Said, A., Martinez, L., and Johnson, K., "A Review of LLM-based Explanations in Recommender Systems," *arXiv preprint arXiv:2411.19576*, 2025.

[8] Silva, K. D., Perera, S. M., and Fernando, A. L., "Sinhala Speech Recognition: Challenges and Opportunities in Low-Resource Language Processing," *International Conference on Asian Language Processing*, pp. 112-119, 2022.

[9] Itransition, "Adopting voice commerce in 2024: Benefits and challenges," *Open Access Government*, Available: https://www.openaccessgovernment.org/adopting-voice-commerce-in-2024-benefits-and-challenges/172027/, Accessed: August 2025.

[10] Google Cloud, "Speech-to-Text API Documentation," Available: https://cloud.google.com/speech-to-text/docs, Accessed: August 2025.

[11] Aravinda, H. W. K., Sirajudeen, R., Karunathilake, S., de Silva, N., Ranathunga, S., and Kaur, R., "SinLlama -- A Large Language Model for Sinhala," in 2025 *Moratuwa Engineering Research Conference (MERCon)*, 2025, doi: 10.1109/MERCon67903.2025.11217094.

[12] A. Pramodya et al., "SinhalaMMLU: A Comprehensive Benchmark for Evaluating Multitask Language Understanding in Sinhala," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025, pp. 32943–32961.

[13] V. Dhananjaya, P. Demotte, S. Ranathunga, and S. Jayasena, "Bertifying Sinhala—A Comprehensive Analysis of Pre-trained Language Models for Sinhala Text Classification," *arXiv preprint arXiv:2208.07864*, 2022.

[14] D. Lakmal, S. Ranathunga, S. Peramuna, and I. Herath, "Word Embedding Evaluation for Sinhala," *in Proceedings of the 12th Language Resources and Evaluation Conference (LREC),* 2020, pp. 1874-1881.

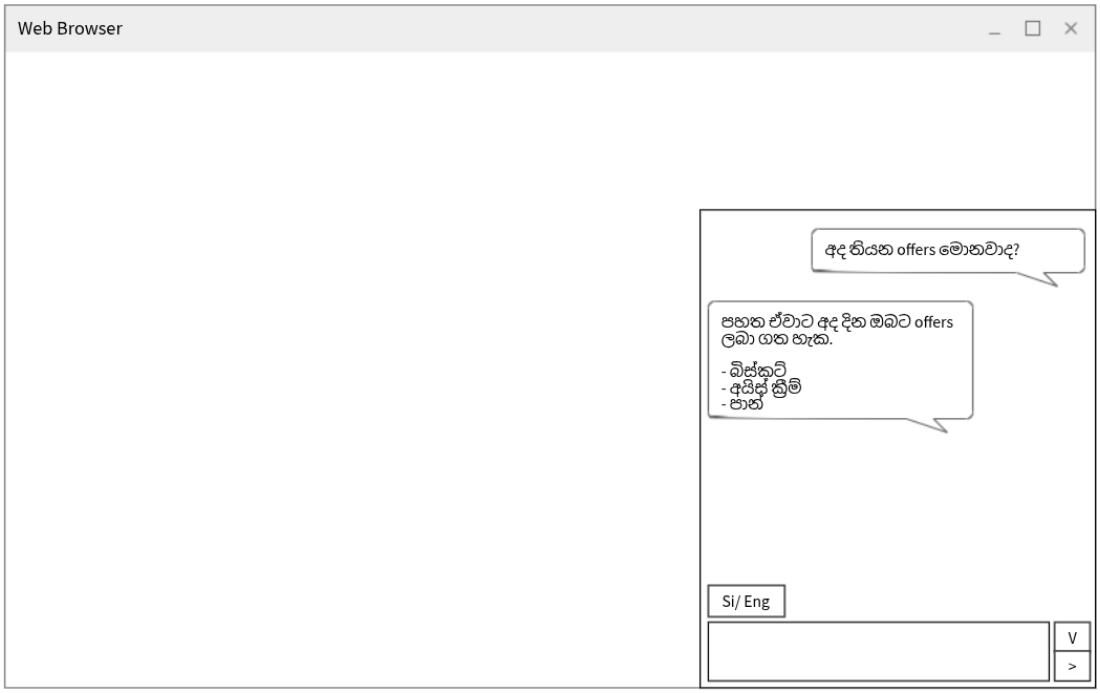
[15] M. T. Ribeiro, S. Singh, and C. Guestrin, " 'Why Should I Trust You?': Explaining the Predictions of Any Classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining,* 2016.

[16] B. Burns et al., "Borg, Omega, and Kubernetes," *Communications of the ACM*, vol. 59, no. 5, pp. 50-57, 2016.

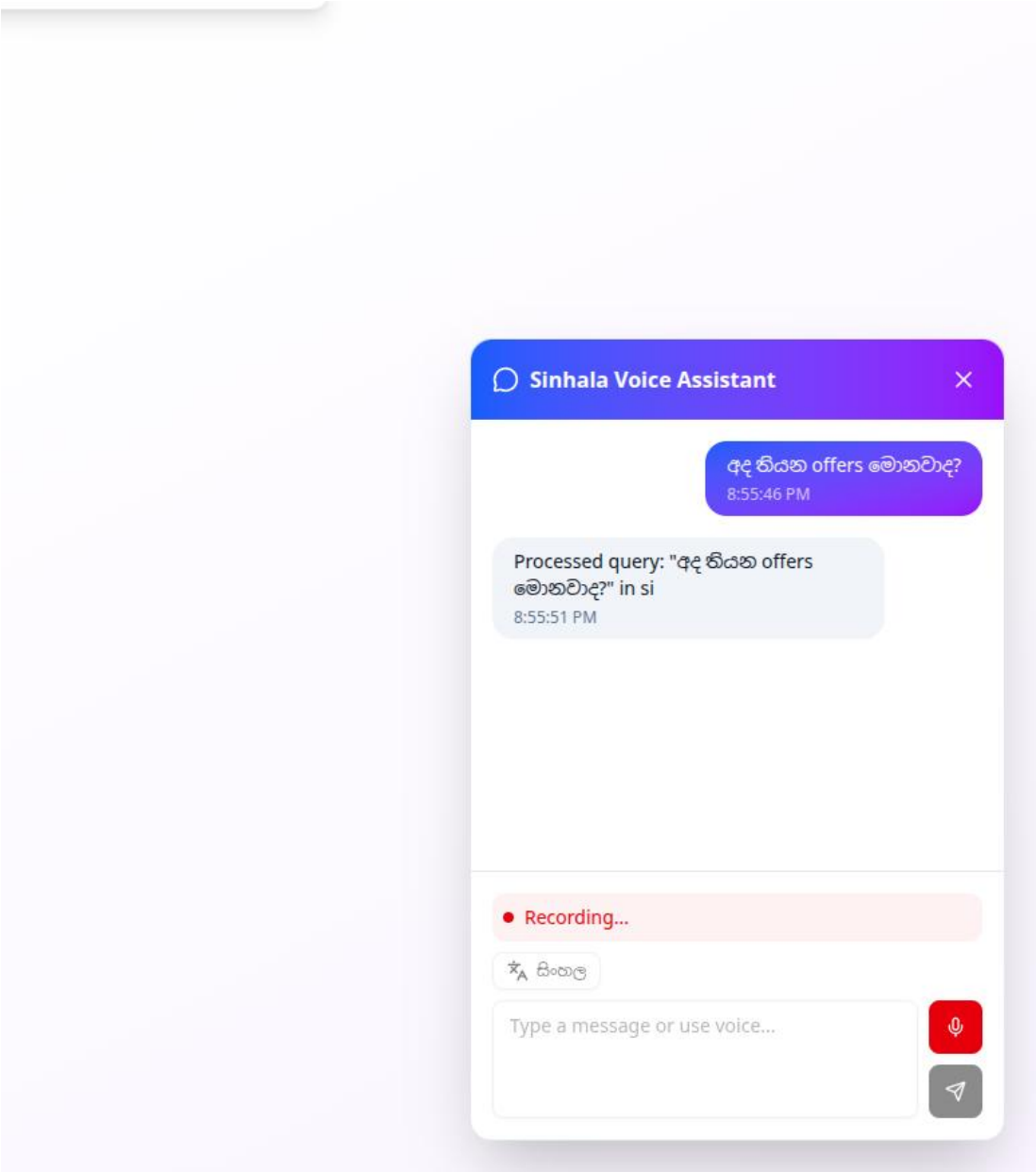
[17] Kubernetes, "Kubernetes Documentation - Microservices," Available: https://kubernetes.io/docs/concepts/services-networking/, Accessed: August 2025.

[18] Mozilla, "Web Speech API Specification," W3C Community Group Draft, Available: https://wicg.github.io/speech-api/, Accessed: August 2025.

6. APPENDICES



Appendix A:Sample wireframe



Appendix B: Sample chat UI